DEBABRATA PANDA

M.S. Artificial Intelligence | Senior Software Engineer | AI/ML | LLM & Agentic AI
Dallas, TX | | debabrata2024panda@gmail.com | Linkedin | GitHub

PROFESSIONAL SUMMARY

Software Engineer and AI/ML professional with 9+ years of industry experience building scalable, production-grade software solutions, currently pursuing an M.S. in Artificial Intelligence at the University of North Texas. Hands-on experience with Python, Java, Generative AI, LLMs, RAG, AI agents, cloud applications, and CI/CD, backed by strong foundations in machine learning, deep learning, and NLP. Focused on building intelligent, reliable, and scalable AI-powered products across Software Engineering, AI/ML, Generative AI, and Applied AI.

EDUCATION

University of North Texas, Denton, TX **2026 - Present**
M.S. in Artificial Intelligence | AREP Research Program
Research interests: LLM applications, AI agents, model evaluation, automation, and reliable AI systems.

NIST, Berhampur, Odisha, India **2011 - 2015**
Bachelor of Technology in Computer Science | CGPA: 7.53/10

TECHNICAL SKILLS

Languages: Python, Java 17, SQL, JavaScript, C/C++ (working knowledge), HTML/CSS
AI / LLM: AI agent design, multi-step orchestration, LLM integration, prompt engineering, GitHub Copilot
Software / Testing: REST APIs, JUnit, Selenium, Robot Framework, Appium, Postman, Swagger, JMeter, Wireshark
Cloud / DevOps: Git, Jenkins, Docker, GitHub Actions, Azure, CI/CD
Data / Systems: MySQL, Linux (RHEL, CentOS, Ubuntu), AIX, TCP/IP, BACnet, MQTT, Modbus
Delivery Tools: JIRA, Azure Boards, Confluence, QTest, TestRail | Certified ScrumMaster (CSM)

PROFESSIONAL EXPERIENCE

Senior Software Quality Engineer - i7 **Jan 2022 - Oct 2024**
Dell Technologies - PowerProtect Data Manager (PPDM) | Bengaluru, India
- Engineered asynchronous Python automation for enterprise backup, recovery, and replication regression testing, reducing test-cycle time 40% and eliminating 800+ hours/year of manual QA effort.
- Designed LLM-powered AI agents and multi-step orchestration workflows for environment provisioning, test-case generation, log triage, and defect tagging, automating approximately 60% of repetitive sprint engineering tasks.
- Architected Jenkins CI/CD pipelines with automated build, test, and deployment gates and Docker-based reproducible test environments, reducing release overhead by approximately 25%.
- Built Java 17/JUnit validation suites across 5+ file systems and 6+ OS platforms and tested REST APIs end-to-end with Postman/Wireshark, contributing to an estimated 30-35% reduction in post-deployment defect remediation effort.

Senior QA Engineer **Feb 2020 - Dec 2021**
75F Smart Solutions - IoT / HVAC Building Automation | Bengaluru, India
- Architected a reusable automation framework using Robot Framework, Python, Selenium, and Appium across web, mobile, and hardware-to-cloud workflows, reducing regression-cycle effort by approximately 50%.
- Developed Python thermal-simulation scripts to validate AI-based predictive HVAC control algorithms across diverse building environments, supporting 15-20% client energy-efficiency gains at go-live.
- Executed BACnet/IoT sensor validation with BACeye and Wireshark and owned API testing from Swagger specifications through Postman contract validation, intercepting hardware and integration defects before field deployment.

Software Engineer (Testing) **Feb 2016 - Jan 2020**
L&T Technology Services - i-BEMS Smart Campus Platform | Bengaluru, India
- Delivered Java-based microservices and REST APIs integrating HVAC, lighting, UPS, sensors, cameras, and occupancy devices across 8+ smart-building use cases.
- Built energy-monitoring dashboards using JavaScript, HTML/CSS, SQL/MySQL, and occupancy data, enabling measurable 10-18% energy-cost reductions for building clients.
- Implemented and tested MQTT, BACnet, and Modbus integrations and pre-commissioning device validation, reducing on-site smart-campus deployment time by an estimated 35-40%.
- Partnered with enterprise clients to define requirements and acceptance criteria and delivered full-stack unit, UI, integration, and performance testing across on-premises and Azure deployments in Agile/TDD sprints.

GRADUATE AI FOCUS

UNT AREP Research Program:

- Conducting applied research on Large Language Models (LLMs), AI agents, model evaluation, and reliable AI systems, with emphasis on practical and production-relevant use cases.
- Building and evaluating AI prototypes, agentic workflows, and automated experimentation pipelines using modern AI/ML techniques and software engineering best practices.
- Investigating model reliability, and contributing to research documentation, experimental analysis, and technical findings with the goal of supporting academic publication and real-world AI system development.